

Introduction to LAE

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Low Altitude Economics

Week 1

Learning Objectives

By the end of this class, you should be able to:

- **Define** the Low Altitude Economy (LAE) in plain English.
- **Identify** the main business opportunities within the LAE.
- **Map** who creates value and who gets paid in the LAE ecosystem.
- **Explain** how Hong Kong's regulatory sandbox can support market development.
- **Use** simple calculations to assess whether an LAE service might be commercially viable.

Why LAE Matters for Hong Kong

Imagine two possible services in Hong Kong:

- A drone carries urgent medical supplies across Victoria Harbour, avoiding road congestion.
- An electric air taxi connects a business district with the airport, reducing door-to-door travel time.

What could this mean for the economy?

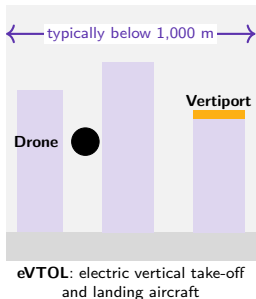
- Low-altitude airspace adds a **third dimension** to the transport network.
- One 2025 study projects that Hong Kong's LAE-related industries could generate approximately **HK\$350 billion in annual value added by 2050**, with supporting services contributing a substantial share.
- **Quick poll (2 min):** "Where is Hong Kong most likely to create value in the low-altitude economy?"

Session 1

Foundations, Definitions & Economic Significance

What is the Low Altitude Economy?

LAE = Economic activity that uses low-altitude airspace, aircraft, infrastructure, and related services.



- **Aircraft:** Drones, helicopters, and eVTOL aircraft.
- **Physical infrastructure:** Vertiports, charging facilities, and maintenance hubs.
- **Digital infrastructure:** Communications, traffic-management software, and data platforms.
- **Applications and services:** Logistics, inspection, emergency response, passenger transport, finance, and insurance.

The exact altitude boundary varies across policy and industry contexts.

How LAE Differs from Traditional Aviation

Recall: LAE covers economic activity using low-altitude airspace and its supporting ecosystem.

Traditional aviation (e.g., Cathay Pacific) operates very differently:

- Uses massive, expensive aircraft
- Departs from giant, centralized airports
- Flies long distances at high altitude

The LAE Shift: Key Business Implications

- **Short-distance, high-frequency operations** inside or between cities
- **Entire ecosystem** required: charging pads, digital traffic software, insurance, and maintenance
- **Analogy:** Think of LAE less like a traditional airline and more like an Uber or broadband internet network for the sky

Result: LAE services have different cost structures, operating models, and regulatory needs.

Jargon Buster: LAE, UAM, and AAM

You will hear these acronyms constantly. Here is what they actually mean:

- **LAE (Low Altitude Economy):** The entire umbrella term we defined earlier. Includes drones, flying taxis, infrastructure, and the software running it all.
- **UAM (Urban Air Mobility):** A specific subset of LAE. This refers *only* to moving people or cargo over short distances inside a crowded city (like Central to Kowloon).
- **AAM (Advanced Air Mobility):** A broader term for using new aircraft to connect cities to suburbs or rural areas.
- **Course focus:** LAE economics starting **HK-first**, then expanding to the **Greater Bay Area (GBA)** and **the rest of China**.

Four Ways to Create Value in the Sky

LAE services must solve a customer problem better than existing alternatives.

1. **Cargo and logistics:** High-priority items for which speed, reliability, or access matters.
2. **Passenger air mobility:** Premium short-distance travel where door-to-door time savings justify the fare.
3. **Industrial inspection:** Inspection of bridges, buildings, power infrastructure, and remote assets with less disruption and risk.
4. **Emergency and public services:** Medical delivery, emergency response, search and rescue, and environmental monitoring.

Managerial question: Which applications create enough value to cover the cost, risk, and regulatory requirements?

Concept Primer: “Value of Time” & “NPV”

1. Value of Time (VOT)

- VOT expresses how much a customer is willing to pay to save travel time.
- It varies across customers, purposes, and situations. Saving 20 minutes may be worth much more for an urgent medical delivery than for a routine trip.

2. Net Present Value (NPV)

- NPV asks whether the future cash generated by a project is sufficient to justify the investment made today.
- A positive NPV suggests that the project may be financially attractive. A negative NPV means that it does not justify the investment under the assumptions used.

Applied Tool: Assessing Market Viability

Step 1: Does the service create value for customers?

$$\text{Value of Time Saved per Trip} = VOT \times \Delta T$$

Step 2: Can the operator capture enough of that value?

$$\text{Annual Operating Profit} = \text{Annual Revenue} - \text{Annual Operating Cost}$$

Step 3: Does the project justify the initial investment?

$$\text{NPV} = -\text{Initial Investment} + \text{Present Value of Future Profits}$$

Core lesson: Time savings support willingness to pay, but fare, demand, costs, and investment determine financial viability.

Back-of-the-Envelope: How Friction Destroys Value

Scenario: A proposed air-taxi service connects HKIA and Central. The customer's time is worth HK\$1,000 per hour.

Case A: Smooth Journey

- Flight time: 10 minutes
- Boarding and waiting: 5 minutes
- Total journey: 15 minutes
- Time saved: 30 minutes = 0.5 hours
- Value of time saved:
 $1000 \times 0.5 = \text{HK\$}500$

Case B: Friction Shock

- Flight time: 10 minutes
- Boarding and waiting: 25 minutes
- Total journey: 35 minutes
- Time saved: 10 minutes ≈ 0.167 hours
- Value of time saved:
 $1000 \times 0.167 \approx \text{HK\$}167$

Takeaway: A fast aircraft does not guarantee a valuable service. Waiting, transfers, and access time can sharply reduce the fare premium that customers will accept.

Illustrative assumptions for classroom analysis; not current market data.

Concept Primer: How Customers Compare Transport Options

When choosing between the MTR, a taxi, or an air taxi, customers consider more than the ticket price.

- **Money cost:** Fare and related charges.
- **Time cost:** In-vehicle time, waiting, boarding, and transfers.
- **Other factors:** Reliability, convenience, comfort, and perceived safety.

Economists combine money and time into a simple measure called **generalized cost**.

Applied Tool: Generalized Cost

The Formula

$$\text{Generalized Cost} = \text{Ticket Price} + VOT \times \text{Travel Time}$$

- The formula converts travel time into a monetary value.
- All else equal, the customer prefers the option with the lower generalized cost.
- A faster mode can charge more, but only up to the value of the time it saves.

Managerial Lesson

Route planners must consider the complete journey, not only the time spent in the aircraft.

Back-of-the-Envelope: Finding the Maximum Ticket Price

Scenario: What is the most a time-sensitive traveller would pay for an air taxi before choosing the Airport Express instead? **Assumption:** VOT = HK\$10 per minute.

Airport Express

- Ticket price: HK\$115
- Journey time: 24 minutes
- $GC_{MTR} = 115 + 10(24)$
- **Generalized cost = HK\$355**

Air Taxi

- Journey time: 8 minutes
- Match the MTR generalized cost:
- $P + 10(8) = 355$
- **Maximum ticket price = HK\$275**

Takeaway: Under these simplified assumptions, HK\$275 makes the traveller indifferent. Actual demand also depends on waiting, access, reliability, comfort, and perceived safety.

Illustrative assumptions for classroom analysis; not current market data.

Hong Kong's Possible Role: The “Service Layer”

If LAE creates substantial value, where might Hong Kong hold an advantage?

- The GBA has significant strengths in aircraft manufacturing, electronics, supply chains, and large-scale testing.
- Hong Kong may contribute high-value supporting services that help LAE businesses operate, finance growth, and expand internationally:
 - **Finance and leasing**
 - **Insurance and risk management**
 - **Legal, certification, and professional services**
 - **Digital platforms and international business development**

This is a strategic proposition to evaluate, not a predetermined outcome.

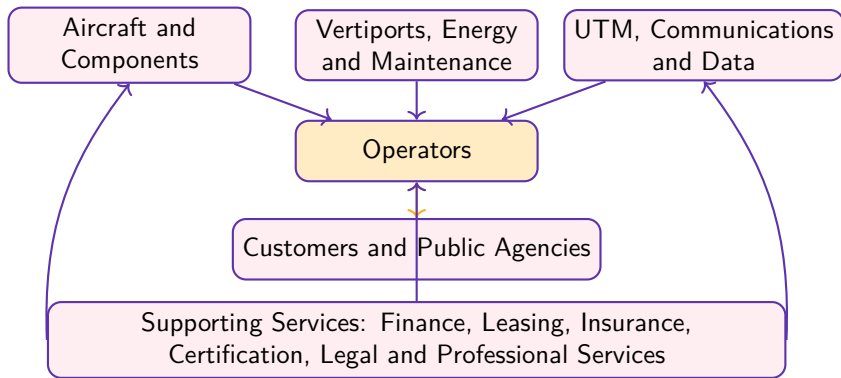
Session 2

Who Makes What? (The Value Chain)

Jargon Buster: The LAE Ecosystem

- **OEM (Original Equipment Manufacturer):** A company that produces the aircraft or key components.
- **Vertiport:** A site designed for eVTOL take-off, landing, passenger handling, charging, and related operations.
- **UTM / U-space:** Digital services and procedures that support the safe and efficient management of unmanned-aircraft operations.
- **Operator:** The organization that provides the transport, logistics, inspection, or public service to the customer.

The LAE Ecosystem – Who Gets Paid?



Managerial question: Which participants create value, bear risk, and have bargaining power?

Applied Tool: How the Ticket Price is Shared

A Simple Price Decomposition

$$\begin{aligned}\text{Ticket Price} = & \text{Aircraft Cost} + \text{Vertiport Fee} + \text{Energy} \\ & + \text{UTM Fee} + \text{Maintenance and Labour} \\ & + \text{Insurance and Overhead} + \text{Operator Margin}\end{aligned}$$

- Different participants may charge usage fees, lease payments, or service fees.
- The operator faces direct demand risk because it must attract customers and fill available capacity.
- Infrastructure and platform providers may face less direct passenger risk, but they still depend on traffic volume and bear investment, technology, and regulatory risks.

Core lesson: Creating value does not guarantee capturing value. Contract design and bargaining power determine how revenue is shared.

Illustrative Unit Economics: Where Might HK\$300 Go?

Scenario: A passenger pays HK\$300 for an air-taxi trip. Assume the fare is allocated as follows.

Payments to Partners

- Vertiport fee: HK\$100
- UTM and digital services: HK\$20
- Insurance and financing: HK\$30
- Aircraft lease allocation: HK\$50
- **Total partner payments: HK\$200**

Operator Economics

- Cash remaining: HK\$100
- Energy: HK\$20
- Maintenance and labour: HK\$50
- **Operator margin: HK\$30**

If all costs remain unchanged, cutting the fare by HK\$50 turns a HK\$30 margin into a HK\$20 loss.

Hypothetical figures for classroom discussion only. Actual costs and contracts may differ substantially.

Stakeholder Conflict: Different Objectives

An **externality** arises when an activity imposes costs or benefits on others that are not fully reflected in the decision-maker's private costs or revenues.

- **Operators:** Want enough flights and passengers to cover fixed costs.
- **Investors:** Want a credible route to approval, revenue, and scale.
- **Government:** Balances innovation, safety, efficiency, and public interest.
- **Communities:** Care about noise, privacy, visual impact, and fair access to benefits.

Managerial challenge: How can a service reach commercial scale while maintaining public acceptance?

Back-of-the-Envelope: How Operating Restrictions Affect Cost

Scenario: Noise concerns lead to a restriction on evening and overnight flights. Assume a fixed aircraft lease cost of HK\$10,000 per day.

Case A: Wider Operating Window

- Flights per day: 50
- Fixed cost per flight:
- $\text{HK\$}10,000 / 50 = \text{HK\$}200$

Case B: Restricted Operating Window

- Flights per day: 20
- Fixed cost per flight:
- $\text{HK\$}10,000 / 20 = \text{HK\$}500$

Takeaway: Public acceptance is not merely a public-relations issue. Operating restrictions can directly affect utilization, unit costs, and commercial viability.

Illustrative assumptions for classroom analysis; not current policy or market data.

Session 3

Hong Kong Policy, "The Sandbox", and Technology

Concept Primer: The Aviation Safety Problem

Why don't we have flying taxis everywhere already?

- Aviation is the safest form of travel in the world because of incredibly strict regulations (often called "Part 25" certification).
- To get a new airplane certified for the public, a company must spend **up to 10 years** and billions of dollars proving it won't crash.
- **The Problem:** Startups building drones do not have billions of dollars. If they have to wait 10 years to fly their first paying customer, they will run out of money and die.

The Solution: Hong Kong's Regulatory Sandbox

A regulatory sandbox allows companies and public agencies to test selected LAE applications under controlled conditions.

- **Test safely:** Trials take place on approved routes and under specified operating conditions.
- **Learn early:** Companies and regulators collect evidence before wider deployment.
- **Reduce uncertainty:** Problems can be identified before very large investments are made.
- **Support regulation:** Trial results help the Government develop appropriate operating and safety requirements.

The sandbox does not remove safety requirements. It provides a controlled pathway for testing and learning.

Source: Hong Kong Transport and Logistics Bureau, Low-altitude Economy Regulatory Sandbox X.

Applied Tool: How a Sandbox May Reduce Entry Barriers

A Simple Framework

$$\text{Entry Barrier} = \text{Investment Cost} + \text{Compliance Cost} \\ + \text{Cost of Delay} + \text{Uncertainty}$$

- **Investment cost:** Aircraft, infrastructure, software, and staff.
- **Compliance cost:** Testing, documentation, professional services, and approvals.
- **Cost of delay:** Cash spent while waiting before wider operation.
- **Uncertainty:** The risk that the technology, business model, or approval pathway may not work as expected.

Core lesson: A sandbox may reduce delay and uncertainty, making a project easier to test and finance. It does not eliminate compliance or safety costs.

Back-of-the-Envelope: The Value of Faster Learning

Scenario: A drone-delivery startup must decide how much capital to commit before it knows whether its service can operate reliably and gain approval.

Without Early Controlled Trials

- Larger upfront commitment
- Slower operational learning
- More uncertainty about routes and requirements
- Greater risk of redesign after investment

With a Regulatory Sandbox

- Smaller staged tests
- Earlier evidence on safety and reliability
- Better information for regulators and investors
- Investment can be linked to milestones

Takeaway: The main economic value of a sandbox is faster, lower-cost learning under controlled conditions.

Working with the Greater Bay Area (GBA)

Hong Kong's LAE development is closely connected with the wider GBA ecosystem.

- **Manufacturing and technology:** Shenzhen and other GBA cities have strong capabilities in drones, electronics, batteries, and supply chains.
- **Hong Kong capabilities:** International finance, insurance, legal services, professional services, logistics, and a complex urban testing environment.
- **Cross-boundary opportunity:** Logistics and public-service applications may create value, but require coordination on routes, customs, data, communications, and safety requirements.
- **Strategic question:** How can Hong Kong and other GBA cities combine complementary strengths rather than duplicate them?

Concept Primer: The "Empty Seat" Problem

How do airlines make cheap tickets?

- A Boeing 777 costs hundreds of millions of dollars. But it has **300+ seats**. By splitting the cost of the fuel and the pilots across 300 people, your ticket to Japan is affordable.

The Flying Taxi Problem:

- An eVTOL flying taxi only has **4 seats**.
- You cannot divide the cost of the pilot and the machine by 300. You divide it by 4.
- To prevent a ticket from costing HK\$5,000, you have to find a different mathematical trick.

Applied Tool: The Unit Cost Equation

The Formula

$$\text{Cost per Passenger} = \text{Variable Cost per Passenger} + \frac{\text{Daily Fixed Cost}}{\text{Flights per Day} \times \text{Seats per Flight} \times \text{Load Factor}}$$

- **Utilization:** How frequently the aircraft flies.
- **Capacity:** How many passenger seats are available.
- **Load factor:** The percentage of seats actually occupied.
- **Core lesson:** Operators need both frequent operations and sufficient demand. Flying often does not help if the aircraft repeatedly flies with empty seats.

Back-of-the-Envelope: Utilization and the Empty-Seat Problem

Assumptions: Fixed aircraft cost = HK\$10,000/day; variable cost = HK\$100/flight; four passenger seats.

Case A: 10 Flights, 50% Load Factor

- Paying passengers:
 $10 \times 4 \times 50\% = 20$
- Fixed cost per passenger: HK\$500
- Variable cost per passenger: HK\$50
- **Total cost: HK\$550 per passenger**

Case B: 25 Flights, 80% Load Factor

- Paying passengers:
 $25 \times 4 \times 80\% = 80$
- Fixed cost per passenger: HK\$125
- Variable cost per passenger:
HK\$31.25
- **Total cost: HK\$156.25 per passenger**

Takeaway: Technology may help, but demand, load factor, charging, maintenance, infrastructure, weather, and regulation all affect achievable utilization.

Illustrative assumptions for classroom analysis; not current market data.

Technology: An Economic Tool

- **Aircraft technology:** Changes speed, range, payload, capacity, reliability, and operating cost.
- **Automation:** May reduce crew requirements and support frequent operations, but must satisfy safety and regulatory requirements.
- **UTM and communications:** Support coordination, monitoring, and scalable operations in complex airspace.
- **Infrastructure:** Charging, maintenance, landing sites, and passenger processing determine actual throughput.

Managerial lesson: A technically impressive aircraft creates commercial value only when the complete operating system is safe, reliable, convenient, and affordable.

In-Class Exercise: Map the Money

Activity: Map the Money in the Low Altitude Economy

- **Format:** Groups of 4–5
- **Scenario:** Choose one:
 - Hong Kong Airport Shuttle route
 - Hospital Medical Delivery route
- **Task:** On paper or in your notes, map the following:
 1. Who are the key players in the LAE Value Chain?
 2. How does the money flow from the customer to each player?
 3. What does each player want? (Incentives and possible conflicts)
 4. How does Hong Kong's Sandbox policy help this business succeed?
- **Timing:** 10 minutes group work + 5 minute quick share-out

Key Takeaways

- LAE combines aircraft, infrastructure, digital systems, operators, customers, and supporting services.
- A service creates value when it saves time, reduces cost or risk, improves access, or enables a mission that existing alternatives cannot perform well.
- Commercial viability depends on willingness to pay, demand, utilization, load factor, operating cost, and investment cost.
- Hong Kong may hold advantages in selected high-value services and in connecting capabilities across the GBA.
- A regulatory sandbox supports controlled testing and faster learning, but does not remove safety or compliance requirements.
- Technology is valuable when it improves the economics of the complete service, not merely the aircraft.

Week 1 Assignment Briefing

Two-page Hong Kong LAE Economic Memo (maximum 800 words)

- **Due:** Beginning of Week 2
- **Task:** Choose one Hong Kong LAE service idea, such as an airport shuttle, medical delivery, inspection service, or logistics route.
- **Requirements:**
 1. Explain the customer problem and why someone would pay for the service.
 2. Estimate a plausible fare or service fee using time savings or another customer benefit.
 3. Estimate annual revenue: $\text{price} \times \text{expected daily transactions} \times \text{operating days}$.
 4. Identify two important barriers, such as demand, cost, infrastructure, regulation, safety, weather, or public acceptance.
 5. Explain how controlled sandbox testing could help address one barrier.
- **Grading focus:** Clear economic logic (40%), transparent and realistic assumptions (30%), policy link (20%), professionalism (10%).

Market Structure: The Fight for the Sky

Question to think about for next week:

If a landing pad on top of a Central skyscraper can only fit two drones at a time... who gets to land there, and who gets pushed out of business?

Thank You + Q&A

Pre-Class Readings Reminder (for Week 2):

- *McKinsey Perspectives on Advanced Air Mobility*
- *NASA UAM Market Study (2018)*

Questions?

(Final 15 minutes reserved for extended Q&A)